

Burnside's lemma and applications in competition problems

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Overview

- Orbit-Stabilizer Theorem
- Burnside's Lemma
- Applications of Burnside's Lemma

What is Burnside's Lemma?

- A lemma found in group theory
- Takes advantage of symmetry in groups to enumerate mathematical objects
- Also a powerful enumerative combinatorial lemma that allows us to solve problems involving counting colorings and arrangements

Preliminary Definitions

Definition

A group G *acts* on a set X if there exists a map $\cdot : G \times X \rightarrow X$ such that $e \cdot x = x$ for all $x \in X$ and $g \cdot (h \cdot x) = (gh) \cdot x$ for all $g, h \in G$ and $x \in X$.

Definition

A *fixed point* of an element $g \in G$ is an element $x \in X$ such that $g \cdot x = x$.

Preliminary Definitions

Definition

For any element $x \in X$, the *stabilizer* of x is denoted by G_x , where G_x is the set of all $g \in G$ such that $g \cdot x = x$.

Definition

For any element $x \in X$, the *orbit* of x in X is denoted by $G \cdot x$, where $G \cdot x$ is the set of all $g \cdot x$ with $g \in G$.

Orbit-Stabilizer Theorem

Theorem

Let G be a finite group that acts on a set X . Let G_x be the stabilizer of an element $x \in X$, and suppose that the orbit $G \cdot x$ of x is finite. Then the index $[G : G_x]$ is finite and equal to $|G \cdot x|$. This means that $|G_x| \cdot |G \cdot x| = |G|$ by Lagrange's Theorem.

Outline of a proof:

- Let $x \in X$, and let G/G_x be the set of all left cosets of G_x
- Define a function $\phi : G \cdot x \rightarrow G/G_x$ by $\phi(g \cdot x) = gG_x$
- Claim: ϕ is a bijection
- Proof: Need to show ϕ is well-defined, injective, and surjective

Orbit-Stabilizer Theorem

- Well-Defined: Suppose that $g \cdot x = h \cdot x$ for some $g, h \in G$; we can show that $h^{-1}g \in G_x$, which implies that $gG_x = hG_x$
- Injective: Suppose that $f(g_1 \cdot x) = f(g_2 \cdot x)$ for distinct $g_1, g_2 \in G$; then $g_1G_x = g_2G_x$, and we can show that $g_1 \cdot x = g_2 \cdot x$
- Surjective: For every $gG_x \in G/G_x$, there exists $g \cdot x \in G \cdot x$ such that $\phi(g \cdot x) = gG_x$
- Thus, ϕ is bijective $\implies |G \cdot x| = |G/G_x| = [G : G_x]$
- By Lagrange's Theorem, $|G| = |G_x| \cdot |G/G_x| = |G_x| \cdot |G \cdot x|$

Burnside's Lemma

Lemma (Burnside's Lemma)

Let G be a finite group that acts on a set X . Let X/G be the set of all orbits of X . For any element $g \in G$, let X^g denote the set of all fixed points of g . Then

$$|X/G| = \frac{1}{|G|} \sum_{g \in G} |X^g|.$$

Burnside's Lemma Proof Outline

Outline of a proof:

- By double-counting, we have that

$$\sum_{g \in G} |X^g| = \sum_{x \in X} |G_x|$$

- By the Orbit-Stabilizer Theorem ($|G_x| \cdot |G \cdot x| = |G|$),

$$\sum_{x \in X} |G_x| = |G| \sum_{x \in X} \frac{1}{|G \cdot x|}$$

- The right-hand side equals $|G| \cdot |X/G|$ by a double counting argument with the contribution of each x in each orbit in X

Coloring the Faces of a Cube

Problem

Suppose that we wanted to count the number of ways to color the faces of a cube with six different colors so that each face is painted exactly one unique color, and two colorings are considered identical if they can be obtained by rotating (but not reflecting) the cube.

Coloring the Faces of a Cube

Solution

- G = group of all rotations of the cube
- $|G| = 24$; 6 ways to choose the top face, 4 ways to choose the front face
 $\implies 6 \cdot 4 = 24$ rotations
- X = set of all colorings of the cube without rotation; $|X| = 6! = 720$
- $g \in G$ not identity rotation $\implies$ no fixed points; any rotation of the cube will give a different coloring
- $g \in G$ identity rotation $\implies X^g = X$; every $x \in X$ is fixed
- Burnside's Lemma $\implies \frac{1}{24} \cdot 720 = \mathbf{30}$ colorings

Arranging Beads on a Necklace

Problem

Suppose that we wanted to count the number of ways to arrange p beads on a necklace, where p is a prime number, such that each bead is either black or white, and two arrangements are considered identical if they can be obtained by rotating (but not reflecting) the necklace.

Arranging Beads on a Necklace

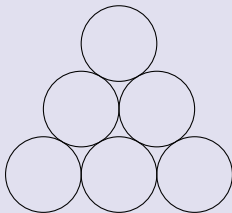
Solution

- G = be the subgroup of symmetric group S_p generated by the cyclic permutation $\sigma = (1\ 2\ \cdots\ p)$; $|G| = p$
- X = set of all arrangements of the necklace without rotation; for each bead, there are 2 choices for its color, so $|X| = 2^p$
- g not identity permutation $\implies$ the arrangements of all black and all white beads are fixed
- Claim: All other $x \in X$ are not fixed
- Proof: Omitted, but uses the fact that p is prime
- Thus, $p - 1$ permutations in G are not the identity, each with two fixed points $\implies$ contributes $2(p - 1)$ to $|X^g|$ sum
- g identity permutation $\implies X^g = X$; every $x \in X$ is fixed, contributes 2^p
- Burnside's Lemma $\implies \frac{1}{p} \cdot (2^p + 2(p - 1)) = \frac{2^p + 2p - 2}{p}$ arrangements

Disk Coloring Counting Contest Problem

Problem (2017 AMC 12B, Problem 13)

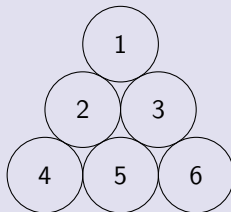
In the figure below, 3 of the 6 disks are to be painted blue, 2 are to be painted red, and 1 is to be painted green. Two paintings that can be obtained from one another by a rotation or a reflection of the entire figure are considered the same. How many different paintings are possible?



Disk Coloring Counting Contest Problem

Solution

- Number the disks as follows:



- G = set of all rotations and reflections of the diagram; $|G| = 6$: 3 rotations (0° , 120° , and 240°) and 3 reflections, over the lines through centers of (1) and (5), (2) and (6), and (3) and (4)
- X = set of all paintings of the diagram without restriction

Disk Coloring Counting Contest Problem

Solution

- 0° rotation: all paintings are fixed points $\implies$ contributes $\binom{6}{3} \cdot \binom{3}{2} = 60$
- 120° and 240° rotations: 0 fixed paintings (one would need the same color for (1), (4), and (6) and the same color for (2), (3), and (5))
- Reflection over line through (1) and (5): green can be placed in 2 ways ((1) or (5)), reds must be placed in (2), (3) or (4), (6) $\implies$ contributes $2 \cdot 2 = 4$
- Other two reflections each contribute 4 fixed paintings
- Burnside's Lemma $\implies \frac{1}{6} \cdot (60 + 4 + 4 + 4) = \mathbf{12}$ paintings

Polynomial Counting Contest Problem

Problem (2010 AIME II, Problem 10)

Find the number of second-degree polynomials $f(x)$ with integer coefficients and integer zeros for which $f(0) = 2010$.

Polynomial Counting Contest Problem

Solution

- Consider second-degree polynomial $f(x)$
- r and s are the zeros of f , and a is the leading coefficient $\implies$
 $f(x) = a(x - r)(x - s)$
- r and s are indistinguishable, so define $G =$ group consisting of identity permutation and permutation that transposes r and s
- $X =$ set of integer triples (a, r, s) such that $f(0) = ars = 2010$

Polynomial Counting Contest Problem

Solution

- Identity permutation $\implies |X| = 4 \cdot 3^4$ fixed points by distributing the prime factors of $2010 = 2 \cdot 3 \cdot 5 \cdot 67$ among a , r , and s in 3^4 ways and 4 ways to assign their signs since we can have a , r , and s be all positive, a and r be negative, r and s be negative, or a and s be negative
- Transposition permutation $\implies 2$ fixed points, namely $r = s = 1$ and $r = s = -1$
- Burnside's Lemma $\implies \frac{1}{2} \cdot (4 \cdot 3^4 + 2) = \mathbf{163}$ such polynomials

Concluding Remarks

- Helpful when we wish to avoid annoying casework or constructive counting
- Can be used to solve problems from coloring the faces of a cube, to counting polynomials with integer coefficients and roots with a specific constant term
- May appear in our lives in ways we do not even expect

We just need a group and a set, and we are all set.

Thank You!

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